



Control of a Nitrogen-Vacancy center as a two-qubit system

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1. Introduction

Nitrogen–vacancy (NV) centers are promising experimental platforms for quantum information processing and quantum sensing. In this work, we study a two-qubit NV system formed by an electron spin coupled to a nuclear spin, where only the electron spin is directly driven by microwave fields:

- Deduce controllability conditions (when the dynamics span $SU(4)$ and when they do not).
- Estimate a Quantum Speed Limit (QSL), an analytic lower bound on the minimal time required to steer the system between two distinguishable states, using Lie-group geometry / Cartan decomposition tools.
- Use numerical optimal control (GRAPE) to design pulses that approach the QSL while keeping control energy/power within reasonable bounds.

2. Model

The evolution of a system is defined through the Schrödinger equation (with $\hbar = 1$) as:

$$\dot{U}(t) = -iH(t)U(t), \quad U(0) = \mathbb{1}, \quad (1)$$

where

$$H = \omega_S S_z + \omega_I I_z + A_{\parallel} S_z \otimes I_z + A_{\perp} S_z \otimes I_x + u S_x + v I_x \quad (2)$$

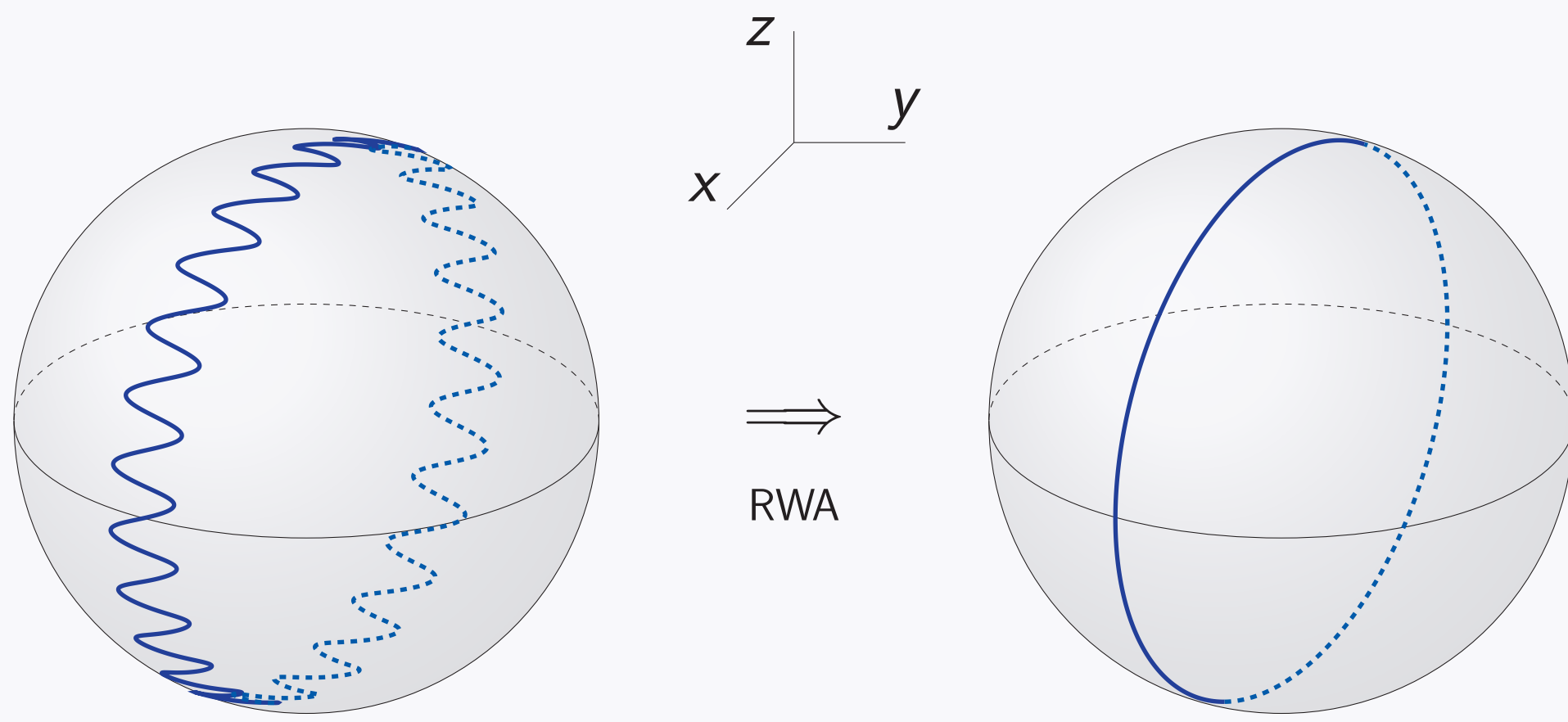
with S_u, I_v are Pauli matrices divided by 2 for $u, v \in \{x, y, z\}$, and

- S : Electron spin
- I : Nuclear spin
- $U(t)$ is the state evolution operator
- $A_{\parallel}, A_{\perp}$ are the parallel and perpendicular hyperfine coupling strengths
- ω_S and ω_I are the zero-field splitting of each individual spin
- u and v are the controls

3. Rotating-wave approximation

Using the reference Hamiltonian $H_R = \omega_S S_z$, the rotating frame Hamiltonian H_{RF} is defined as $U(t)^\dagger (H - H_R) U(t)$ with $U(t) = e^{-itH_R}$. Applying the Corollary 1 in [1], we obtain:

$$H_{RWA} = \omega_I I_z + A_{\parallel} S_z I_z + A_{\perp} S_z I_x + \frac{u_0}{2} (\cos(\phi_S) S_x - \sin(\phi_S) S_y) + v I_x. \quad (3)$$

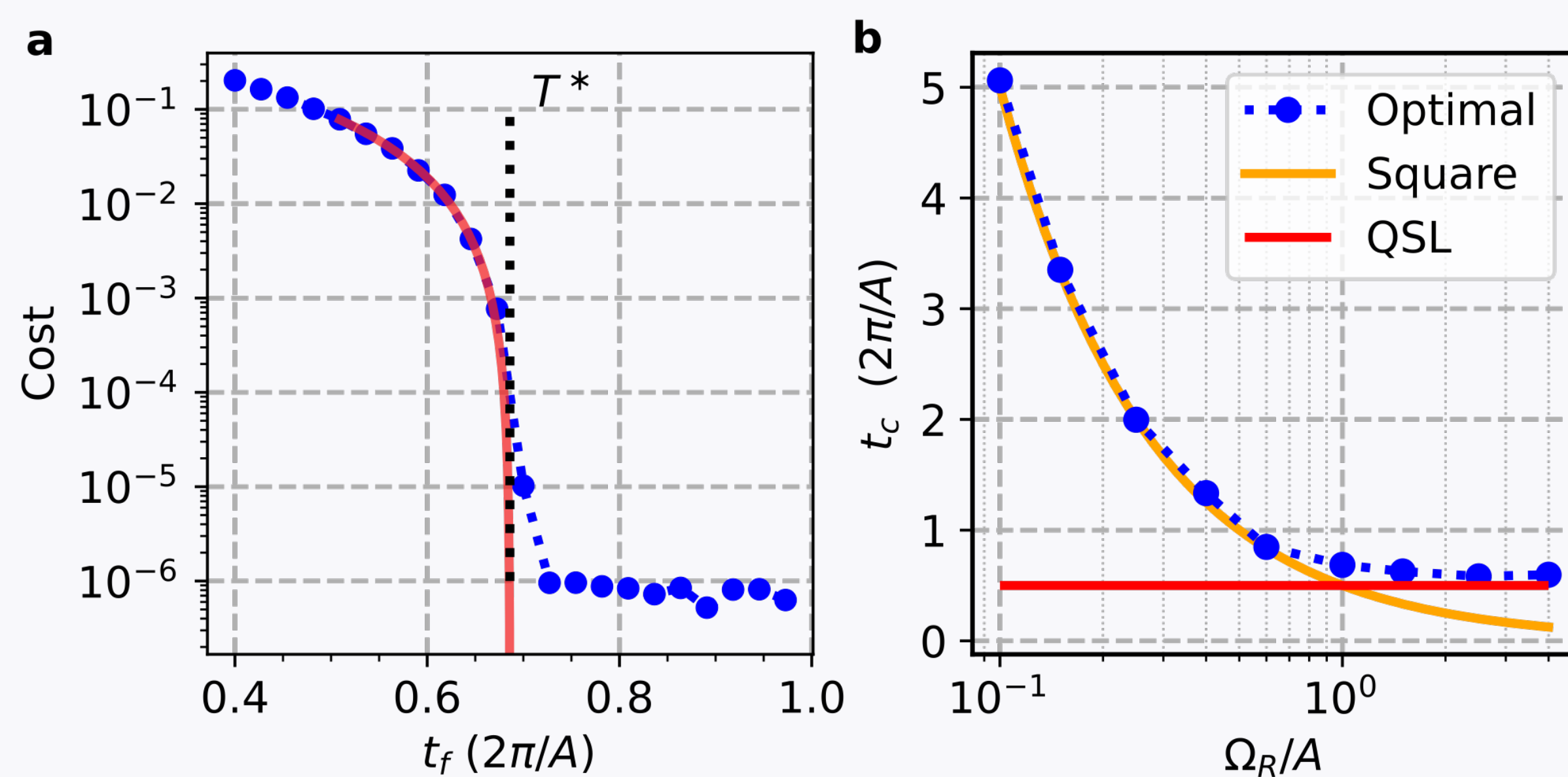


The convergence error for a time $T > 0$ is bounded by

$$\|U_{RF} - U_{RWA}\| \leq \frac{\Omega}{4\omega_S} \left(1 + 2T(\omega_I + A_{\parallel} + A_{\perp}) + \frac{\Omega}{2\omega_S} (2\omega_S + 1) \right).$$

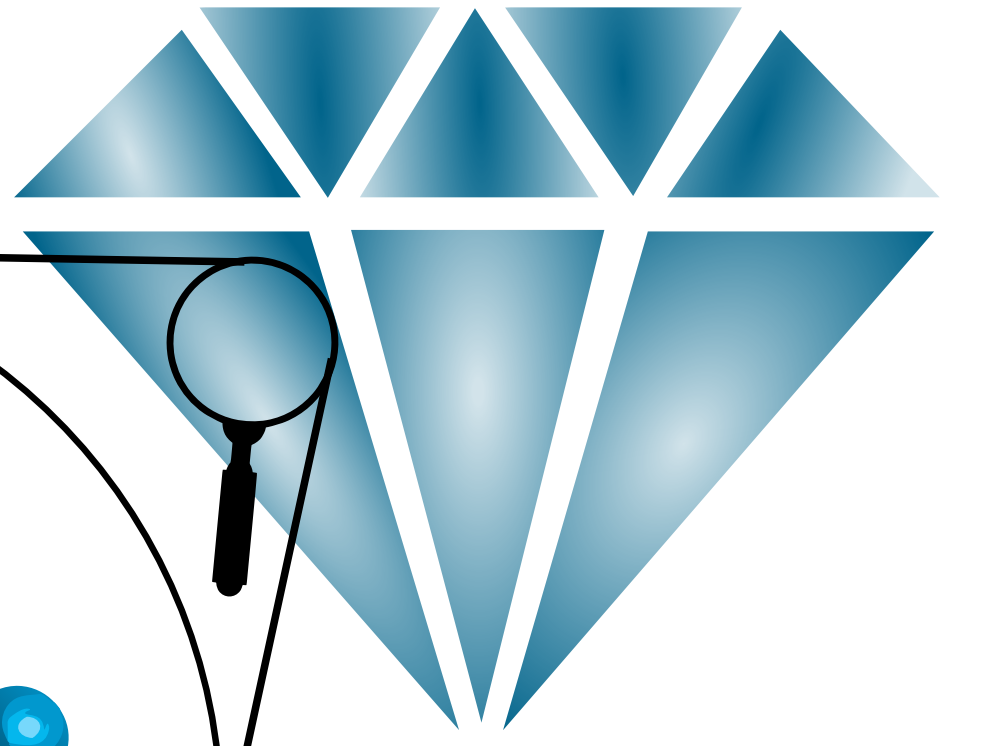
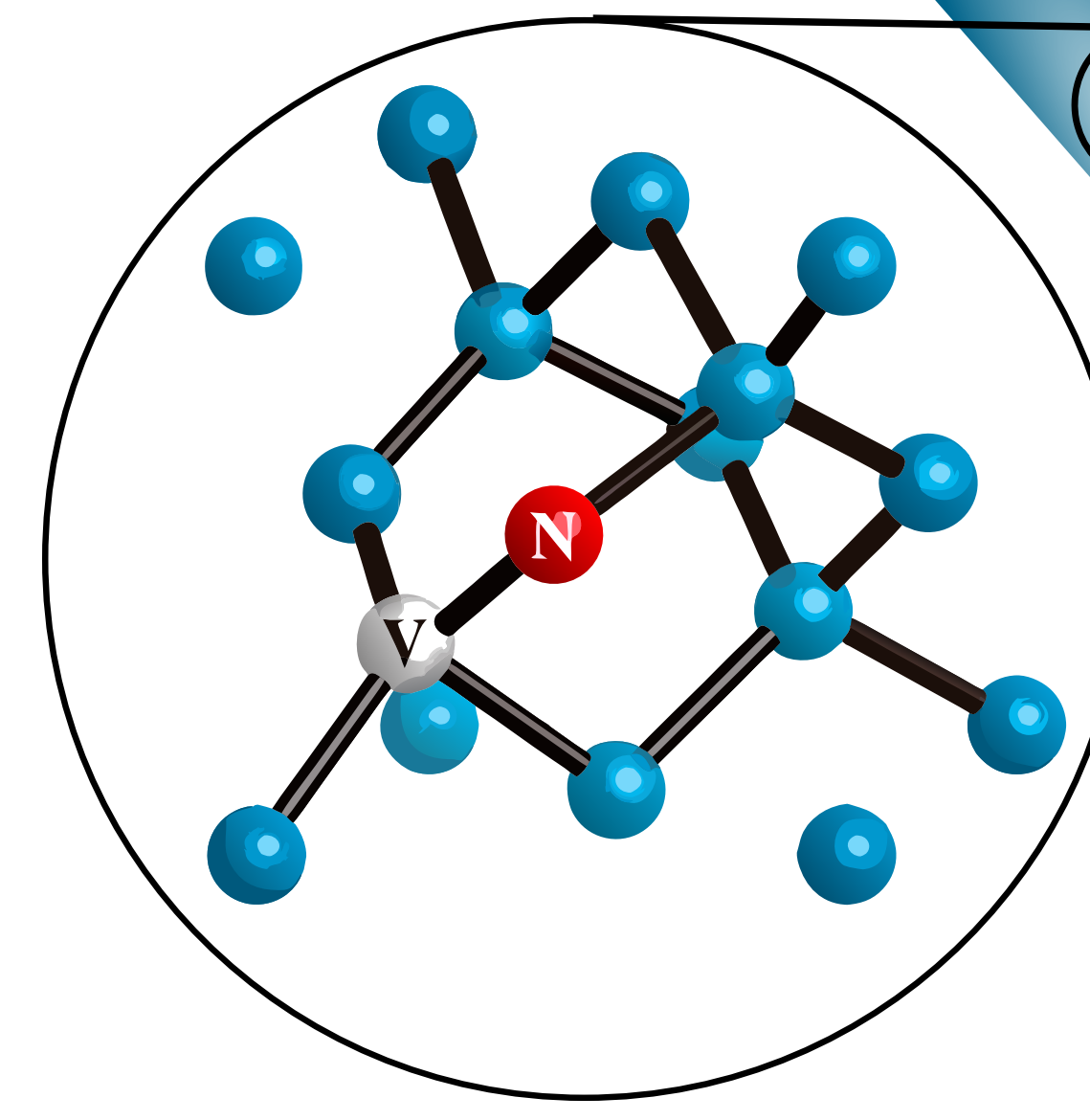
6. Numerical results

- Gradient Ascent Pulse Engineering (GRAPE) algorithm;
- Using $u_0 = 2\Omega_R$, where Ω_R is the Rabi frequency;
- $\dot{U}(t) = -i [A_{\parallel} S_z I_z + \Omega_R (\cos(\phi(t)) S_x - \sin(\phi(t)) S_y)] U(t)$;
- CROT;
- $C = 1 - \text{Fidelity} = 1 - \frac{1}{4} \sum_{i,j \in \{0,1\}} \|\langle i, j | U_t^\dagger U_{t_f} | i, j \rangle\|^2$;
- For each Ω_R , optimizations are performed for several final times t_f . The critical time T^* is extracted by fitting the exponential-decay region of the cost curve.

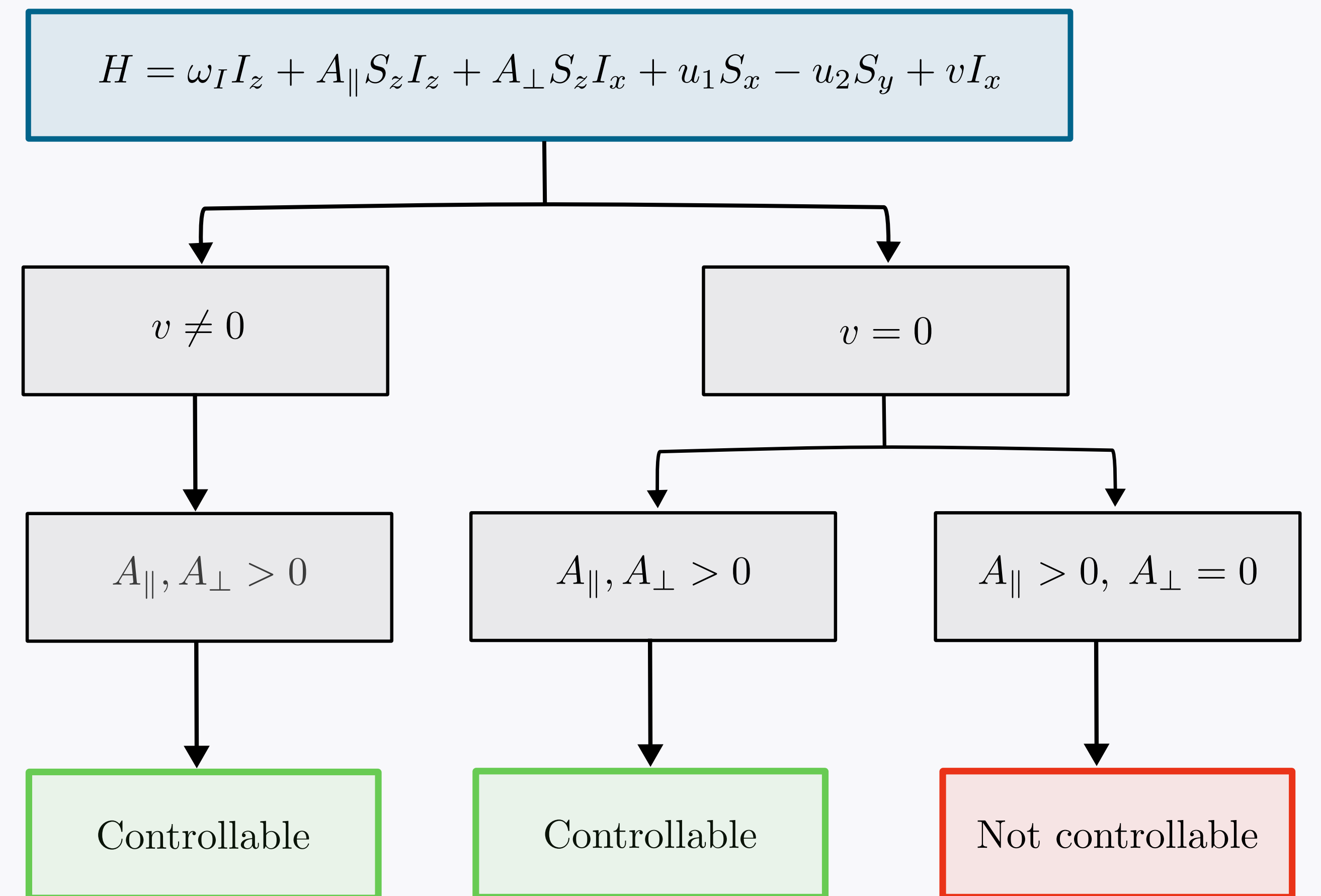


7. Conclusion

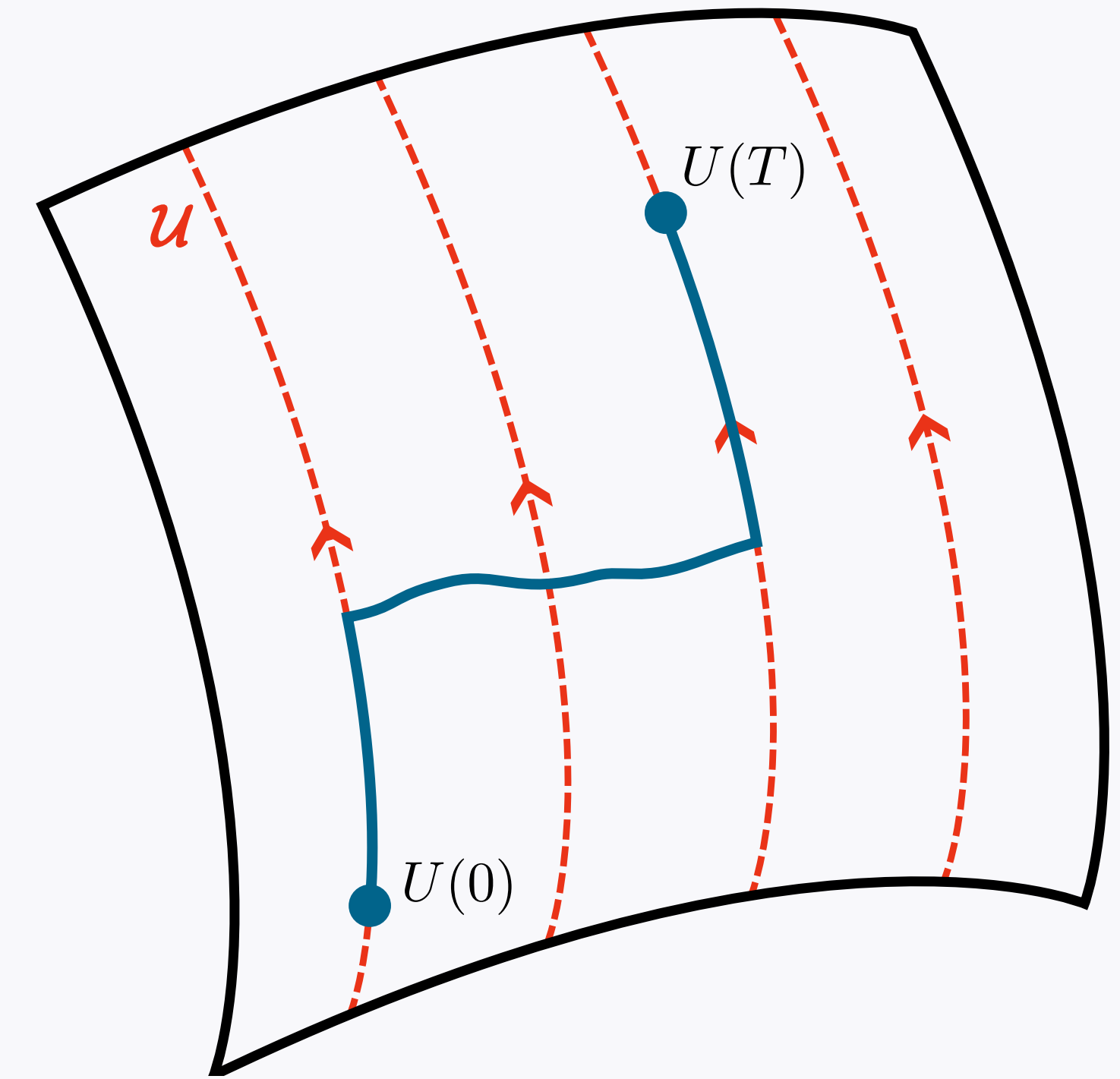
We applied control theory tools to study system controllability and to estimate the QSL for generation of unitary rotations. Furthermore, the numerical results reached the theoretical estimations.



4. Controllability



5. Quantum speed limit



For Hamiltonian (3) with $A_{\perp} = 0$, using the Theorem 2 in [2], we obtain:

- G the group with the associated Lie algebra $\mathfrak{g} = \text{Lie}\{iS_z I_z, iS_x, iS_y\}$,
- K be a subgroup with the associated Lie algebra $\mathfrak{k} = \text{Lie}\{iS_x, iS_y\}$,
- $G/K \cong SU(2)$ is a homogeneous space, which has the direction of the control flow.
- Consider $\langle X, Y \rangle_G = \text{Tr}(X^\dagger Y)$ and $\mathfrak{p} = \mathfrak{k}^\perp = \text{Lie}\{iS_z I_z, iS_y I_z, iS_z I_z\}$
- $\mathfrak{g}$ can be decomposed along Cartan decomposition, $\mathfrak{g} = \mathfrak{p} \oplus \mathfrak{k}$,
- $\{iS_z I_z\} \subset \mathfrak{p}$ is a maximal Abelian subalgebra of rank one.
- The minimal time to steer the system from $\mathbb{1}$ to U_f is the smallest time $t > 0$ such that $U_f = Q_1 \exp(\pm t A_{\parallel} S_z I_z) Q_2$ with $Q_1, Q_2 \in K$. Since the drift is 2π periodic, the worst case scenario is when the system needs to make a π rotation, which gives the minimal time equal to $\pi/A_{\parallel}$.

8. Future work

- Extension to higher spins: Spin 1 coupled with spin $\frac{1}{2}$ and other variations.
- Construct the optimal controls for this type of system.
- Comparison between direct and indirect algorithms.